

# Assignment-01

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Sec : 13

CSE320

Ans. to the ques. no. 1

|          | D.MAC | S.MAC | D.IP | S.IP | D.Port | S.Port | Data | T <sub>2</sub> |
|----------|-------|-------|------|------|--------|--------|------|----------------|
| Router 1 | H     | E     | 11   | 6    | 443    | 49543  |      |                |

$$\frac{10510801x0501}{1} = \frac{2}{1} = 2$$

$\frac{1}{9d} \cdot \frac{0007}{24600} = 829C+5$

## Ans. to the ques. no. 2

Given,

$$T = 10 \text{ ns} \\ = 10 \times 10^{-9} \text{ s}$$

We know,

$$f = \frac{4}{T} = \frac{4}{10 \times 10^{-9}} = 4 \times 10^8 \text{ Hz} \quad \text{✓}$$

$$\therefore T = \frac{1}{f} = 2.5 \times 10^{-9} \text{ s}$$

That is, the signal suffers from distortion.

### Ans. to the ques. no. 3

Given,

$$BW = 20 \text{ kHz}$$

$$f_{\min} = 25 \text{ kHz}$$

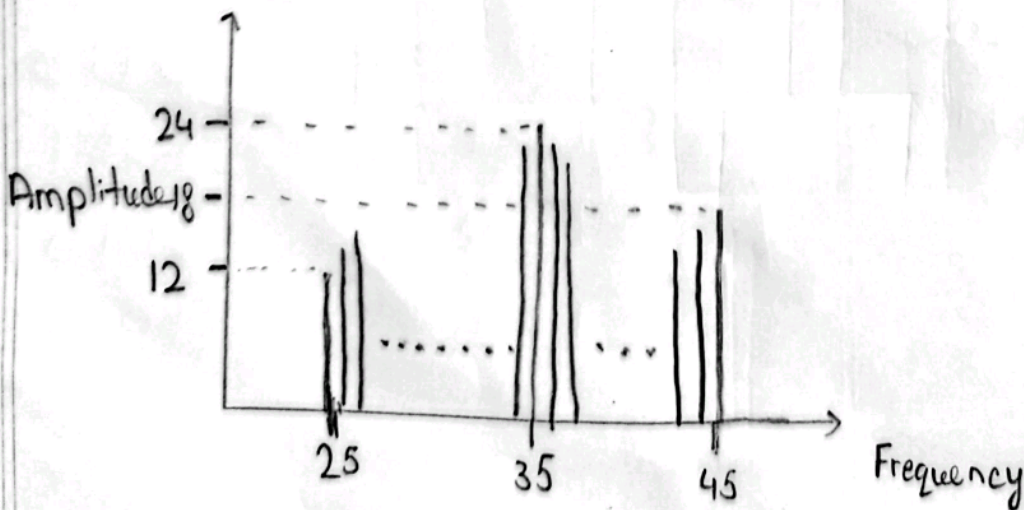
We know,

$$BW = f_{\text{high}} - f_{\text{low}}$$

$$\text{or, } f_{\text{high}} = BW + f_{\text{low}}$$

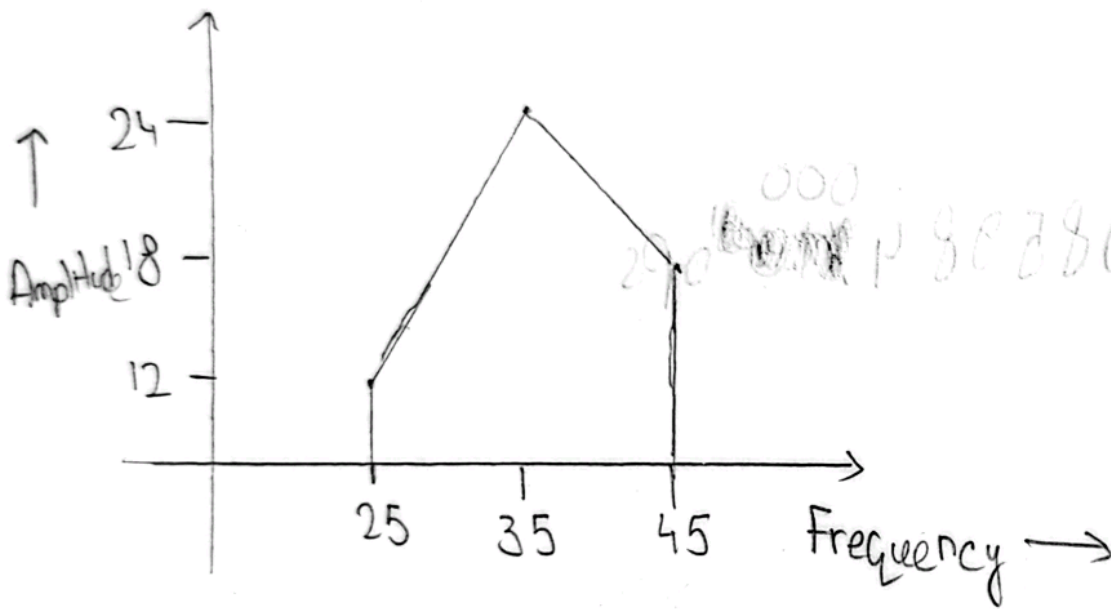
$$= 20 + 25$$

$$= 45 \text{ kHz}$$



$$f_{\text{mid}} = \frac{45 + 25}{2}$$
$$= 35 \text{ Hz}$$

Ans. to the ques.no. 4



Ans. to the ques. no. 5

a

We know,

$$s = vt$$

$$\text{or, } v = \frac{s}{t} = \frac{1920 \times 1080 \times 60 \times 24}{1} \\ = 2985984000 \text{ bps}$$

Principle of 20A

b

From (a) we get,

$$C = 2985984000 \text{ bps} \\ L = 32$$

We know,

$$C = 2B \log_2 L$$

$$\text{or, } B = \frac{C}{2 \log_2 L} = \frac{2985984000}{2 \log_2 32} = 298598400 \text{ Hz}$$



C

$$C' = 0.2 \times 1.2$$

$$= 3583180800 \text{ bps}$$

$$B = 298598400 \text{ Hz}$$

We know,

$$C' = B \log_2 (1 + \text{SNR})$$

$$\text{or, } \log_2 (1 + \text{SNR}) = \frac{C'}{B}$$

$$\text{or, } 1 + \text{SNR} = 2^{12}$$

$$\text{or, } \text{SNR} = 4096 - 1$$
$$= 4095$$

$$\text{or, } \log_2 (1 + \text{SNR}) = 12$$

$$(2 \times 8 - 0.5) = 89$$

$$(41 - 0.5) =$$

$$40.5 =$$

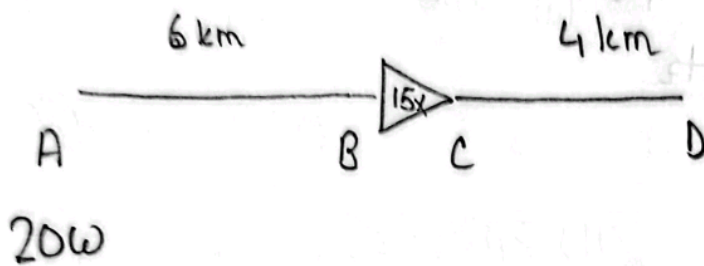
$$40.5 = 12 \times 8 = 96$$

$$(12 \times 8 - 0.5) = 95.5$$

$$(30 - 1.5) =$$

$$28.5 =$$

Ans. to the ques. no. 6



Attenuation =  $3W/km$

$$P_B = (20 - 3 \times 6)$$

$$= (20 - 18)$$

$$= 2W$$

$$P_C = 15 \times 2 = 30W$$

$$P_D = (30 - 3 \times 4)$$

$$= (30 - 12)$$

$$= 18W$$



$$\therefore A_{AB} = 10 \log_{10} \frac{P_B}{P_A}$$

$$= 10 \log_{10} \left( \frac{2}{20} \right)$$

$$= -10 \text{ dB}$$

$$\therefore A_{BC} = 10 \log_{10} \frac{P_C}{P_B}$$

$$= 10 \log_{10} \left( \frac{30}{2} \right)$$

$$= 11.76 \text{ dB}$$

$$\therefore A_{CD} = 10 \log_{10} \frac{P_D}{P_C}$$

$$= 10 \log_{10} \left( \frac{18}{30} \right)$$

$$= -2.218 \text{ dB}$$

$$\begin{aligned} \therefore \text{Total attenuation } A_D &= A_{AB} + A_{BC} + A_{CD} \\ &= (-10 + 11.76 - 2.218) \\ &= -0.458 \text{ dB} \end{aligned}$$

Ans. to the ques. no. 7

Given,

$$P_s = 3.6 \text{ mW} = 3.6 \times 10^{-3} \text{ W}$$

$$\text{Distance} = 10 \text{ km}$$

$$\therefore \text{Total attenuation} = -0.45 \times 10 = -4.5 \text{ dB}$$

We know,

$$AH = 10 \log_{10} \frac{P_D}{P_s}$$

$$\text{or, } -4.5 = 10 \log_{10} \frac{P_D}{3.6 \times 10^{-3}}$$

$$\text{or, } \frac{P_D}{3.6 \times 10^{-3}} = 10^{\frac{-4.5}{10}} = 0.354$$

$$\text{or, } P_D = 0.354 \times 3.6 \times 10^{-3}$$

$$= 1.2744 \times 10^{-3} \text{ W}$$

$$= 1.2744 \text{ mW}$$

Ans. to the ques. no. 8

a

We know,

$$S = vt$$

$$\text{or, } v = \frac{S}{t} = \frac{1.35 \times 10^8 \times 8}{5 \times 60} = 3.6 \times 10^6 \text{ bps} = 3.6 \text{ Mbps}$$

$$B = 100 \text{ kHz} = 10^5 \text{ Hz}$$

Here,

$$v = 0.75 \text{ C}$$

$$\text{or, } C = \frac{v}{0.75} = \frac{3.6 \times 10^6}{0.75} = 4.8 \times 10^6 \text{ bps} = 4.8 \text{ Mbps}$$

b

We know,

$$C = B \log_2 (1 + \text{SNR})$$

$$\text{or, } (1 + \text{SNR}) = 2^{\frac{C}{B}} = 2^{\frac{4.8 \times 10^6}{10^5}} = 2.814 \times 10^{14}$$

$$\text{or, } \text{SNR} = 2.814 \times 10^{14} - 1$$

$$= 2.814 \times 10^{14}$$

C

We know,

$$C = 2B \log_2 L$$

$$\text{or, } \log_2 L = \frac{C}{2B}$$

$$= \frac{3.6 \times 10^6}{2 \times 10^5} = 18$$

$$\text{or, } L = 2^{18} = 262144$$

Ans. to the ques. no. 9

Given,

$$\text{Queuing time} = 2 \mu s = 2 \times 10^{-6} s$$

$$\text{Processing " } = 1 \mu s = 1 \times 10^{-6} s$$

Now,

$$\text{Propagation ~~time~~ <sup>delay</sup>} = \frac{2000 \times 10^3}{2 \times 10^8} = 0.01 s$$

$$\text{Transmission delay} = \frac{5 \times 10^6 \times 8}{5 \times 10^6} = 8 s$$

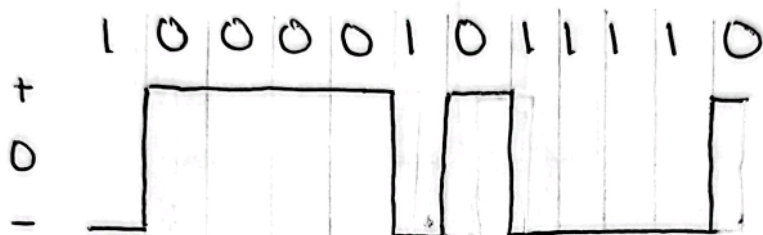
$$\begin{aligned} \therefore \text{Latency} &= 2 \times 10^{-6} + 1 \times 10^{-6} + 0.01 + 8 \\ &= 8.010003 s \end{aligned}$$

Dominant component is transmission delay

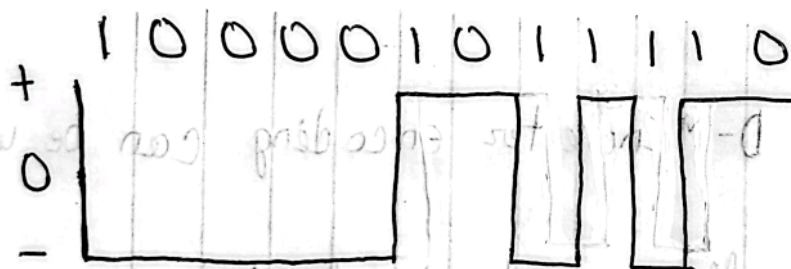
Processing ~~time~~ <sup>delay</sup> is negligible.

## Ans. to the ques. no. 10

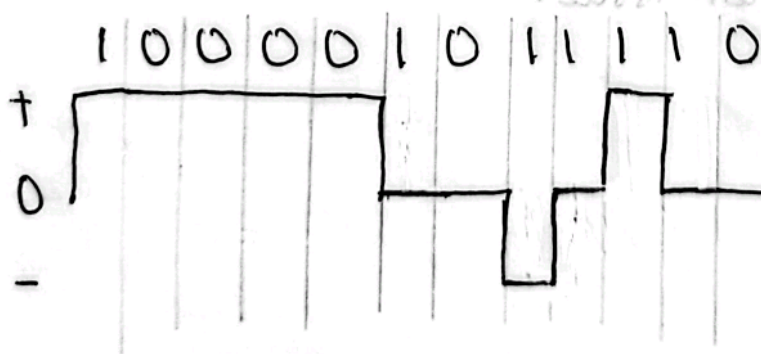
NRZ-L



NRZ-I



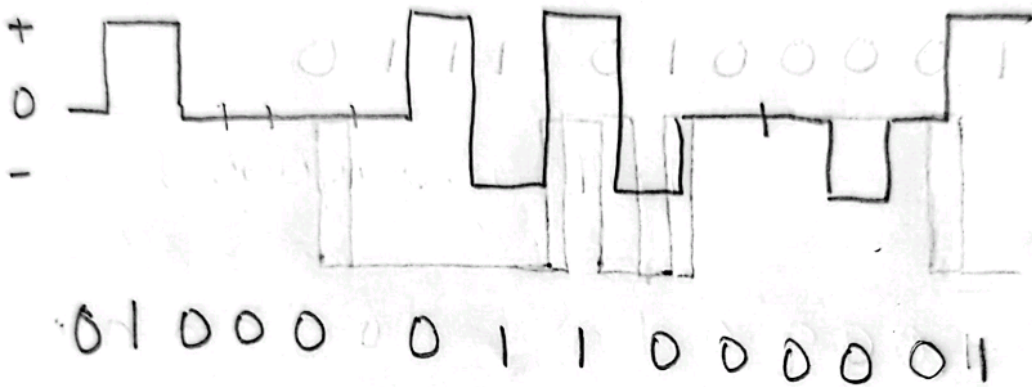
MLT-3





Ans. to the ques. no. 11

a



The number is = 010000110000001

b

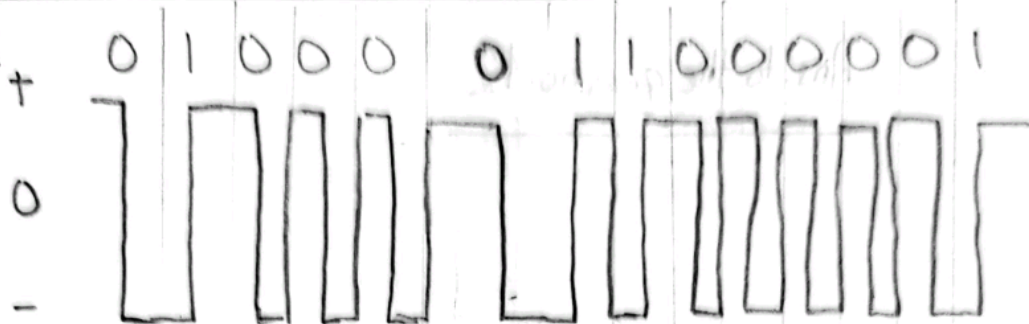
0111101000001

Manchester and D-Manchester encoding can be used as they do not suffer any kind of baseline wandering

and DC component issue.



MAN



The signal level is coded above the zero level.

DMAN



Here, DMAN is better choice due to its immunity to polarity inversion and better noise resistance.

Ans. to the ques. no. 12

VAM

a

Its using AMI line coding scheme. The signal is -

110000000000101

VAM

b

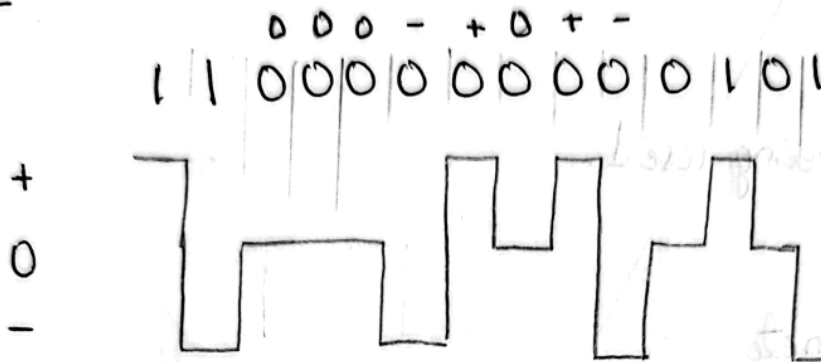
This signal suffers from long sequence of zeros. AMI

has baseline wandering issue which causes loss of

synchronization due to long sequence of zeros.

C

B8ZS



Ans. to the ques no. 13

1 1 0 1 0 1 0 0 0 0 1 1

~~0000~~

~~1000~~

~~0001~~

1001

0010

1010

0011

1011

0100

~~1100~~

0101

~~1101~~

0110

~~1110~~

~~0111~~

~~1111~~

Data

Encoded

000

0010

001

0011

010

0100

011

0101

100

0110

101

1001

110

1010

111

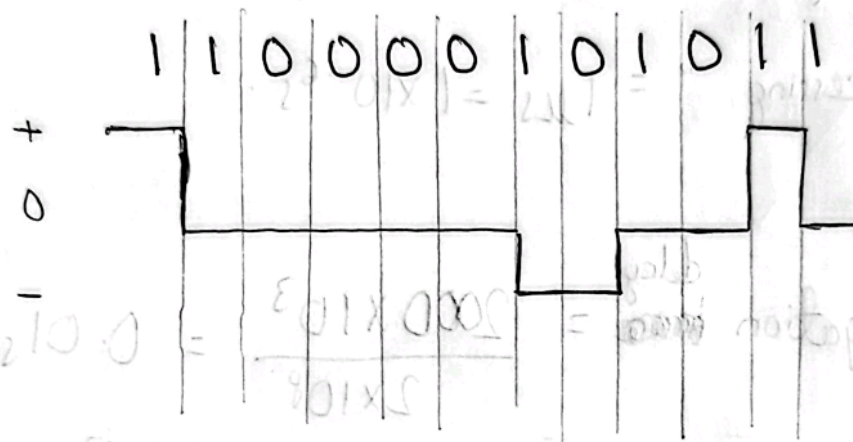
1011



Ans. to the ques. no. 14

a MLT-3

b)



c HDB3

